

Question

1. What is a conditional statement in programming?

A **conditional statement** in programming is a decision-making statement that executes different blocks of code depending on whether a condition is **true** or **false**.

Definition

A conditional statement checks a condition and performs an action based on the result.

Example

If a student's marks are greater than or equal to 40, the student passes; otherwise, the student fails.

Question

- 2. What does an if statement do?
- It is a decision making statement in c/c++ that checks whether a condition is true or false.

Question

- 3. What is the purpose of an else block?
- The else block is used to execute the block of code when the condition is false.

Question

- 4. When is an else block executed?
- Else block is executed when the condition in the corresponding if statement is false.

Question

- 5. What is a Boolean expression?
- A Boolean statement is an expression that evaluates to only two possible values
- True
- False
-

Question

- 6. Write the syntax of a simple if statement
- If (condition)
- {
 //statements to be executed
- }.

Question

- 7. What operator is commonly used to test equality?
- Equality operator (==) is used to test whether two values are equal in c/c++

Question

- 8. What does the modulus (%) operator return?
- Modulus operator returns the remainder when one integer is divided by another .

Question

- 9. How can you check if a number is even?
- A number will be even if it leaves a remainder of 0 when divided by 2.

Question

- 10. How can you check if a number is odd?
- A number is said to be odd if it leaves a remainder when divided by 2.

Question

- 11. What is the purpose of else-if?
- The main purpose is to check
Multiple conditions one after another.

Question

- 12. What is a nested if statement?
- A nested if statement is an if statement placed inside another if statement .

Question

- 13. Can an if statement exist without an else?
- Yes if statement can exist without an else statement.

Question

- 14. What is the output when a false condition is tested in an if statement without else?
- When a **false condition** is tested in an if statement **without an else block**, the statements inside the if block are **not executed**. The program simply skips the if block and continues with the next statement.

Question

- 15. What is the role of braces {} in C/C++ conditionals?
- Braces {} are used to **group multiple statements into a single block** in conditional statements such as if, else, and else if.

Question

- 16. What does > mean?
- The > (**greater than**) operator is a **relational operator** used to compare two values.

Question

- 17. What does < mean?
- The < (**less than**) operator is a **relational operator** used to compare two values.

Question

- 18. What does $\geq$ mean?
- The $\geq$ (**greater than or equal to**) operator is a **relational operator** used to compare two values.

Question

- 19. What does $\leq$ mean?
- The $\leq$ (less than or equal to) operator is a relational operator used to compare two values.

Question

- 20. What does != mean?
- The != (not equal to) operator is a relational operator used to compare two values.

Question

- 21. What is a switch statement?
- A **switch statement** is a decision-making statement that allows a program to choose one block of code from multiple alternatives based on the value of an expression.

Question

- 22. What is a case label in a switch statement?
- A **case label** is a value inside a switch statement that specifies a particular option to be matched with the value of the switch expression.

Question

- 23. Why is break used in switch?
- The **break** statement is used to **terminate the execution of a switch case** and exit the switch block.

Question

- 24. What happens if break is omitted?
- If the **break** statement is omitted, the program continues executing the statements of the next case(s) even if their labels do not match. This is called **fall-through**.

Question

- 25. What is the default case in switch?
- The **default case** is executed when none of the case labels match the value of the switch expression.

Question

- 26. Can switch compare strings in C++?
- **No**, a switch statement cannot directly compare **strings** in standard C++.
- A switch statement works only with:
- int
- char
- enum
- Other integral types

Question

- 27. Which statement is better for many constant choices: if-else or switch?
- For **many constant choices**, a **switch statement** is generally better than an if-else ladder.

Question

- 28. How do you test whether a value is positive?
- A value is **positive** if it is **greater than zero**.

Question

- 29. How do you test whether a value is negative?
- A value is **negative** if it is **less than zero**.

Question

- 30. How do you test whether a value is zero?
- A value is **zero** if it is equal to **0**.

Question

- 31. Write a condition to check if age is at least 18

age >= 18

Question

- 32. Write a condition to check if marks are above 50.
- `marks > 50`

Question

- 33. What is decision making in programming?
- **Decision making** is the process by which a program chooses between different actions based on whether a condition is true or false.
- It is done using conditional statements such as:
 - if
 - if-else
 - else if
 - switch

Question

- 34. Can nested if statements have multiple levels?
- **Yes**, nested if statements can have **multiple levels**. An if statement can be placed inside another if statement, and this nesting can continue as needed.

Question

- 35. What is logical AND (&&)?
- The **logical AND operator (&&)** is used to combine two or more conditions.
- It returns:
- **True (1)** if all conditions are true.
- **False (0)** if any one condition is false.

Question

- 36. What is logical OR (||)?
- The **logical OR operator (||)** is used to combine two or more conditions.

Question

- 37. What is logical NOT (!)?
- The **logical NOT operator (!)** is used to **reverse** the result of a condition.

Question

- 38. How do you check if a year is divisible by 4?
- `year % 4 == 0`

Question

- 39. What is a leap year?
- A **leap year** is a year that has **366 days** instead of **365 days**.

Question

40. `int a = 10, b = 20;`

- `if(a > b)`
- `{`
- `cout << "Largest = " << a;`
- `}`
- `else`
- `{`
- `cout << "Largest = " << b;`
- `}`

Question

- 41. How do you find the largest of two numbers?
- `int a = 10, b = 20;`
- `if(a > b)`
- `{`
- `cout << "Largest = " << a;`
- `}`
- `else`
- `{`
- `cout << "Largest = " << b;`
- `}`

Question

- 42. How do you find the largest of three numbers?
- `int a = 10, b = 25, c = 15;`
- `if(a >= b && a >= c)`
- `{`
- `cout << "Largest = " << a;`
- `}`
- `else if(b >= a && b >= c)`
- `{`
- `cout << "Largest = " << b;`
- `}`
- `else`
- `{`
- `cout << "Largest = " << c;`
- `}`

Question

- 43. What is an if-else-if ladder?
- An **if-else-if ladder** is a conditional structure used to check **multiple conditions** one after another.
- If the first condition is true, its block executes.
- Otherwise, the next else if condition is checked.
- If none of the conditions are true, the else block executes.

Question

- 44. What is the difference between if and switch?
- The if statement is used to test complex conditions and ranges, while the switch statement is used to select one block of code from multiple constant choices. if is more flexible, whereas switch is more organized for many fixed options.

Question

- 45. Can switch use ranges directly?
- **No**, a switch statement in C/C++ cannot directly use ranges such as 1-10 or 50-100.

Question

- 46. What is fall-through in switch?
- **Fall-through** occurs when a case in a switch statement does **not have a break statement**. The program continues executing the next case(s) even if their labels do not match.

Question

- 47. What data type is usually used for conditions?
- The **Boolean (bool)** data type is usually used for conditions in C++.
- A Boolean value can have only two values:
- true
- false

Question

- 48. Why are conditional statements important?
- Conditional statements are important because they allow a program to **make decisions** and perform different actions based on different conditions.

Question

- 49. Give one real-life example of a conditional statement.
- A real-life example of a **conditional statement** is:
- **If it rains, then I will carry an umbrella.**
- **Condition:** It rains.
- **Action/Result:** I will carry an umbrella.

Question

- 50. Name four conditional structures covered.
- The four conditional structures covered in C/C++ are:
 - Simple if
 - if-else
 - if-else-if ladder
 - switch statement